

# AI Delegation Boundary

**A pull request for agent authority  
from agent capability  
to governed operational authority**

WebMCP tools → challenge gate → human judgment → exact apply

Technical briefing deck

August 2026

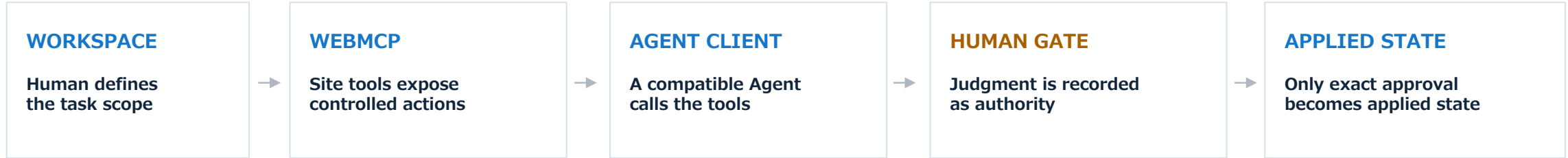
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Web app state, WebMCP capability surface, human judgment, exact approval, and verification boundaries

**SOLIFAN**



# Agent capability is no longer the control boundary



## INITIAL BUILD RESULT

**5**

Normal tools

**6**

After approval

**32**

Tests passed

**0**

AI backend

## CONTROL LIMITATION

- A web app alone should not imply Agent authority.
- A ready state is not the same as human approval.
- An Agent must not create the decision that authorizes itself.

The focus moves from “Can an agent act?” to “Which human-approved state can be applied, based on which challenge?”

# Separate Agent capability from application authority

| Layer                                                                                 | Adopted responsibility                                                           | Boundary evidence                  |
|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------|
| 01 Web application                                                                    | Owns boundary, Guardrails, human judgments, approval, and applied state          | UI state + revision history        |
| 02 WebMCP runtime                                                                     | Makes site tools discoverable to a compatible Agent client                       | document.modelContext tools        |
| 03 Agent client                                                                       | Inspects, proposes, challenges, reviews, and applies only when capability exists | Tool calls and outputs             |
| 04 Human authority                                                                    | Scopes the work, resolves challenges, approves exact revisions                   | Decision + timestamp + fingerprint |
| 05 Static deployment                                                                  | Serves the workspace without an AI backend or hidden model service               | GitHub Pages build artifact        |
| Design rule: WebMCP gives the Agent access; the web app keeps the authority boundary. |                                                                                  |                                    |

# Five normal tools; sixth exists only after exact approval

1

**inspect\_delegation\_workspace**  
Read task, current revision, approval, applied state, and tool availability

4

**review\_delegation\_revision**  
Run Guardrails, challenge gate, and regression checks

2

**propose\_boundary\_revision**  
Create a candidate revision for the scoped work

5

**inspect\_revision\_history**  
Read preserved revision history, statuses, and review results

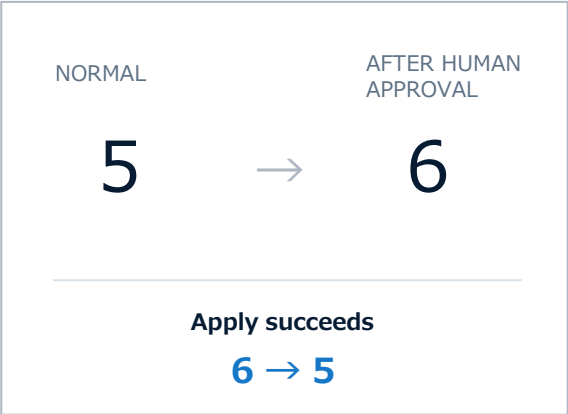
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**challenge\_boundary\_revision**  
Generate a concrete scenario that may break the boundary

6

**apply\_approved\_revision**  
Available only after exact human approval; removed after apply

## DYNAMIC CAPABILITY

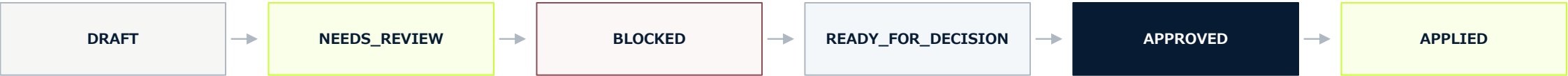


The Agent can call tools, but cannot create the human approval that unlocks the sixth tool.

# Human scopes work, Agent proposes/challenges, human judges, review may loop



## REVISION STATE MACHINE



● BLOCKED → HUMAN EDIT → DRAFT → FRESH CHALLENGE    Replaced non-APPLIED revisions are preserved as SUPERSEDED.

## AUTHORITY RULES

### Human-only scope first

Agent tools that change, challenge, or review authority are locked until the work is scoped.

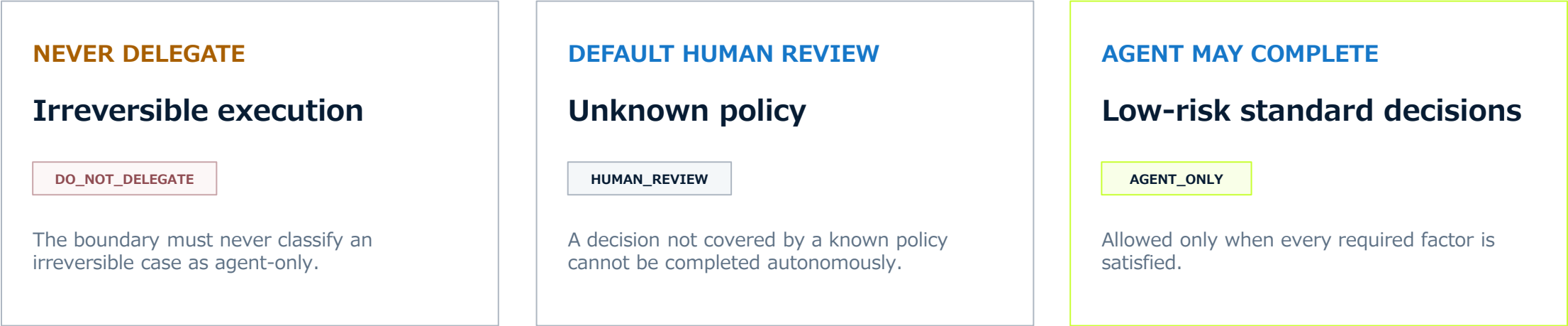
### READY\_FOR\_DECISION is not approval

A passing review is an input to human judgment; it is not itself a decision.

### Apply is exact

Only the revision and fingerprint approved by the human can be applied.

# Encode what can be delegated without losing human control



## FACTOR CONTRACT



Default boundary: irreversible execution uses the highest rule priority; unknown policy routes to HUMAN\_REVIEW.

# A boundary cannot be approved without trying to break it

## GATE CONTRACT

|    |                       |                                                                        |
|----|-----------------------|------------------------------------------------------------------------|
| 01 | Boundary changes      | A revision changes the authority rule for the scoped task.             |
| 02 | Fresh Agent Challenge | The Agent must create a concrete scenario against that exact boundary. |
| 03 | Human judgment        | A human resolves the challenge; the result becomes a Known Decision.   |
| 04 | Review                | Guardrails and Known Decisions are checked before approval.            |

## GUARDRAIL INVARIANT CHECK

| Guardrails                                                                                                                    | Challenge                      | Judgment                 |
|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------|--------------------------|
| checked outside Agent-selected cases                                                                                          | required after boundary change | required before approval |
| <div><div>NO CHALLENGE → NO APPROVAL</div><div>Prior challenge results are invalidated when the boundary changes.</div></div> |                                |                          |

Guardrails are checked independently of Agent-selected Challenges; non-negotiable violations block READY\_FOR\_DECISION.

# Human judgments become tests; exact approval stays separate

## E2E FLOW EXAMPLES → REGRESSION MEMORY

| Scenario                                           | Human resolution | Future revision effect                                                     |
|----------------------------------------------------|------------------|----------------------------------------------------------------------------|
| 01    Example: MEDIUM evidence + low impact refund | KEEP_HUMAN       | If a later boundary yields AGENT_ONLY, review fails as regression.         |
| 02    Example: HIGH evidence + standard refund     | ALLOW_AGENT      | That human judgment is preserved as a regression test for the same facts.  |
| 03    Exact approval record                        | NOT MEMORY       | Handled by the Exact Approval contract on Slide 9, not by Known Decisions. |

Every human override becomes a test before it becomes a rule.



# Apply only the approved revision and fingerprint



## APPROVAL CONTRACT

### Exact state only

Approval is bound to the same reviewable state, not to chat context or a label.

### READY is not approval

A review result can be READY\_FOR\_DECISION and still require a human decision.

### No capability residue

After application, apply\_approved\_revision is no longer available.

**Human approval changes the Agent capability surface — temporarily, and only for the exact approved state.**

# Built as a static WebMCP-enabled application

## STACK

- React 19
- TypeScript
- Vite
- Vitest
- WebMCP via document.modelContext
- GitHub Pages

## VERIFICATION

32

automated tests

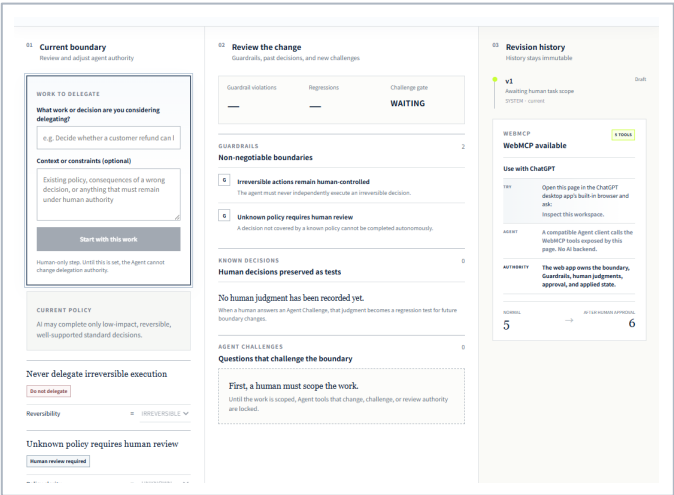
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test files

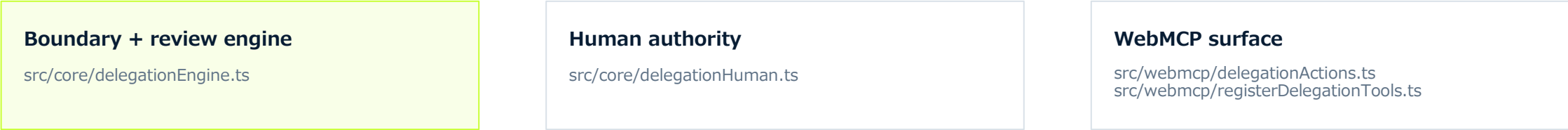
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AI backend

Production build passed. 32 tests cover the engine, human authority, exact approval/apply contract, and WebMCP actions.



## IMPLEMENTATION MAP



Live app: <https://solidfdn.github.io/decision-patch-webmcp/> | Repository: <https://github.com/solidfdn/decision-patch-webmcp>

# A pull request model for agent authority

## IMPLEMENTED

- ✓ WebMCP site tools and dynamic capability surface
- ✓ Human task scope before authority changes
- ✓ Guardrail invariants and mandatory fresh challenge
- ✓ Known Decisions as regression tests
- ✓ Exact-state Human Approval before apply
- ✓ Apply tool appears only after approval and disappears after use

## KNOWN CONSTRAINTS / NEXT HARDENING

- Demo state is browser-based; production would need server persistence and tenant isolation.
- No enterprise identity, retention, SIEM, or audit storage integration yet.
- No runtime model router or hosted AI backend by design.
- Challenge scenarios demonstrate the boundary model; customer-specific policies would be configured later.
- Submission build is a working WebMCP prototype, not a production authorization system.

**Result: agent authority is treated like production code — propose, challenge, test, review, approve, then apply.**